Vadose Zone

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Streamflow problem set (#5) deadline pushed to Tuesday
- Literature review this Thursday

522 Term Project Lit. Review

1. Preliminary water balance equation
2. Review academic/grey literature for studies relating to any component of the water balance in the region of your control volume (or a related region):
   - What did they do?
   - What did they find?
   - What were the limitations?
3. How will this previous work inform your hydrologic analysis and what gaps remain that you aim to address?
   - Could use their methods
   - Could use their water balance (component) as a starting point or point of validation

Not looking for multiple pages here, a page or two is fine!

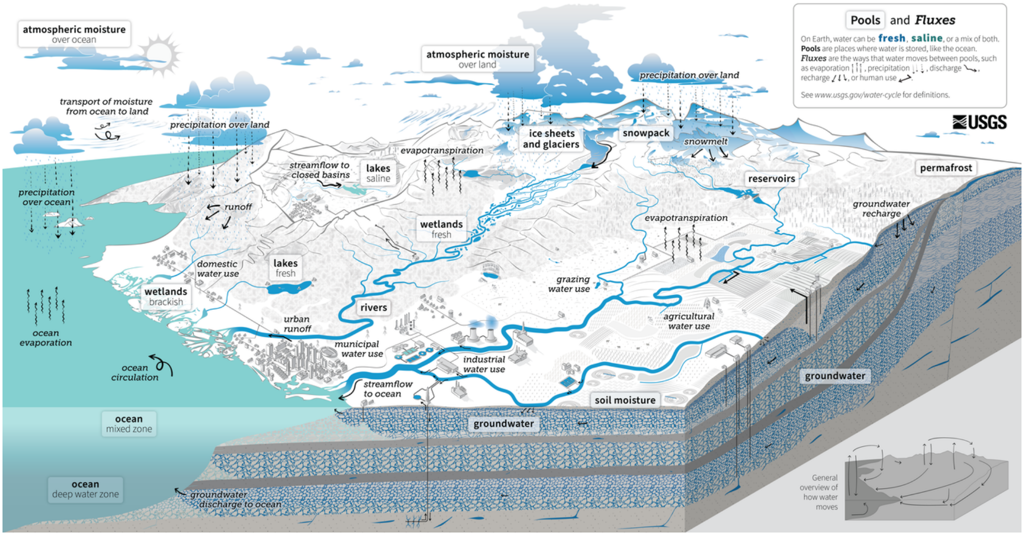
Job Opportunities

Marcy Litvak (eddy covariance guest lecture) is hiring the following:

1. Full-time tower technician (1-year, up to 3-years)
2. Data manager to process flux tower data
3. Technician
4. Student
5. Postdoc working on scaling fluxes to the watershed scale using machine learning

Please reach out to her for more information: mlitvak@unm.edu

Water Cycle



atmospheric moisture
over ocean
transport of moisture
from ocean to land
precipitation over land
atmospheric moisture
over land
precipitation over land
Pools and Fluxes
On Earth, water can be fresh, saline, or a mix of both.
Pools are places where water is stored, like the ocean.
Fluxes are the ways that water moves between pools, such as evaporation, precipitation, discharge, recharge, or human use.
See www.usgs.gov/water-cycle for definitions.
USGS
ice sheets
and glaciers
snowpack
snowmelt
evapotranspiration
streamflow to
closed basins
lakes
saline
permafrost
reservoirs
precipitation
over ocean
runoff
wetlands
fresh
evapotranspiration
groundwater
recharge
domestic
water use
lakes
fresh
wetlands
brackish
grazing
water use
rivers
agricultural
water use
urban
runoff
ocean
evaporation
municipal
water use
industrial
water use
ocean
circulation
streamflow
to ocean
groundwater
soil moisture
groundwater
ocean
mixed zone
ocean
deep water zone
groundwater
discharge to ocean
General
overview of
how water
moves

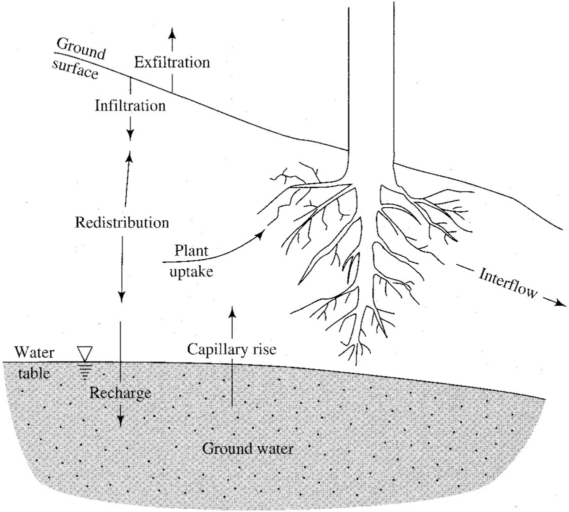
Learning Objectives

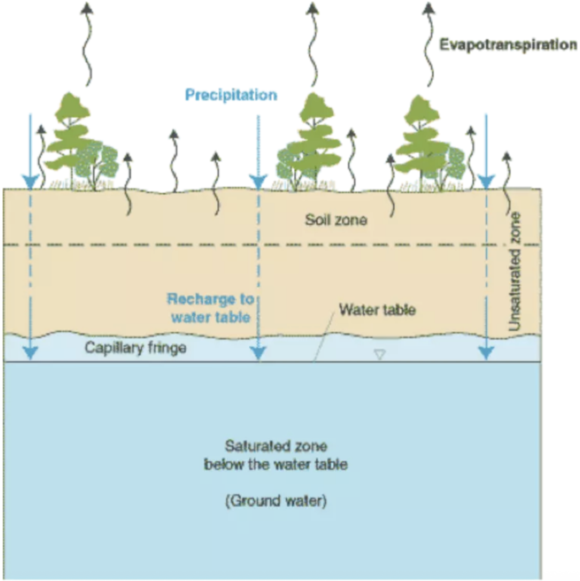
1. Material properties of the subsurface
2. Water content and pressure
3. Infiltration
4. Redistribution

Introduction

Roughly 76% of all precipitation on the land surface **infiltrates**, which is the movement of water from the surface into the subsurface.

Here this water is **redistributed**, or moved around due to pressure gradients, and can be taken up by plant roots (transpiration), become subsurface stormflow (interflow), or **percolate** down to the saturated zone (groundwater recharge).





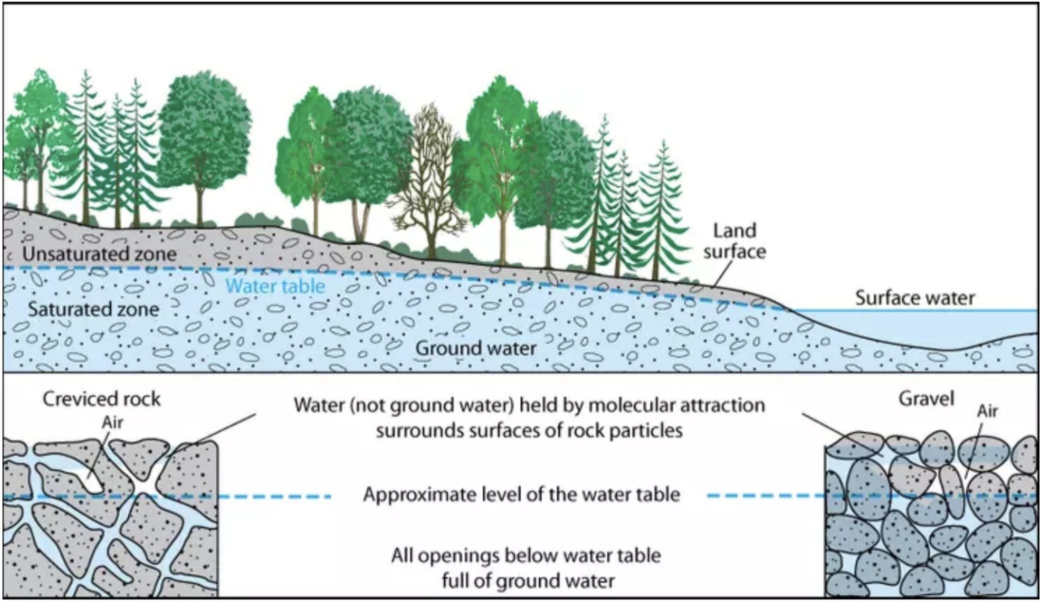
Vadose (unsaturated) zone: void space contains water and air.

Soil zone: uppermost portion of the subsurface that is most biotically active (i.e., plant roots, organic matter, mineral layers). Can be unsaturated or saturated.

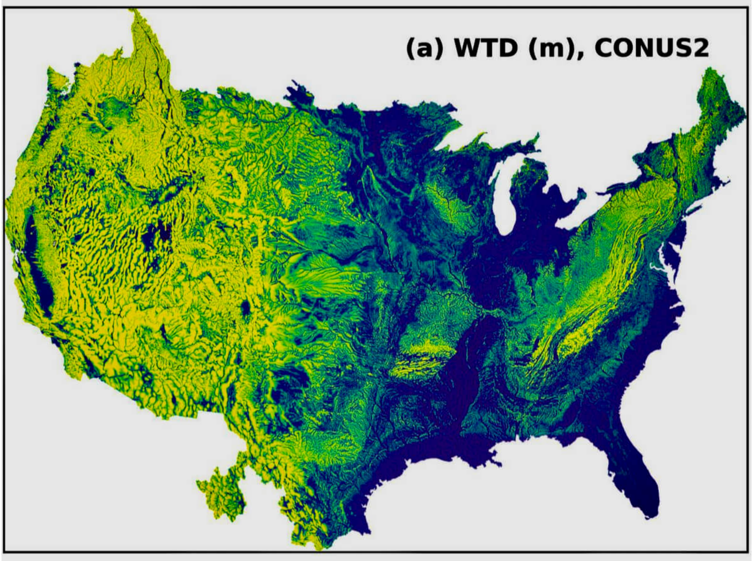
Phreatic (saturated) zone: void space completely filled with water.

Water table (▽): upper "surface" of saturated zone.

Capillary fringe: zone of upward movement from saturated to vadose zone.


Land surface
Unsaturated zone
Water table
Saturated zone
Surface water
Ground water
Creviced rock
Air
Water (not ground water) held by molecular attraction surrounds surfaces of rock particles
Gravel
Air
Approximate level of the water table
All openings below water table full of ground water

Thickness of Vadose Zone



Yang et al. (2023)

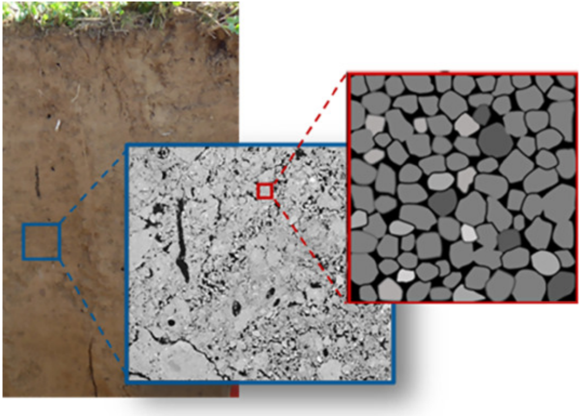
yellow = thicker vadose zone

blue = shallower WT thinner VZ

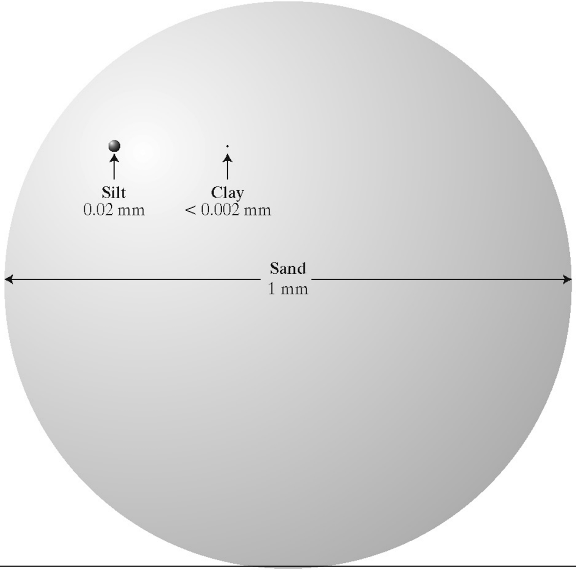
controls: climate elevation agriculture geology

Porous Medium Properties

Porous medium: matrix of individual solid grains (organic or inorganic) with interconnected pore spaces that contain varying proportions of air/water.

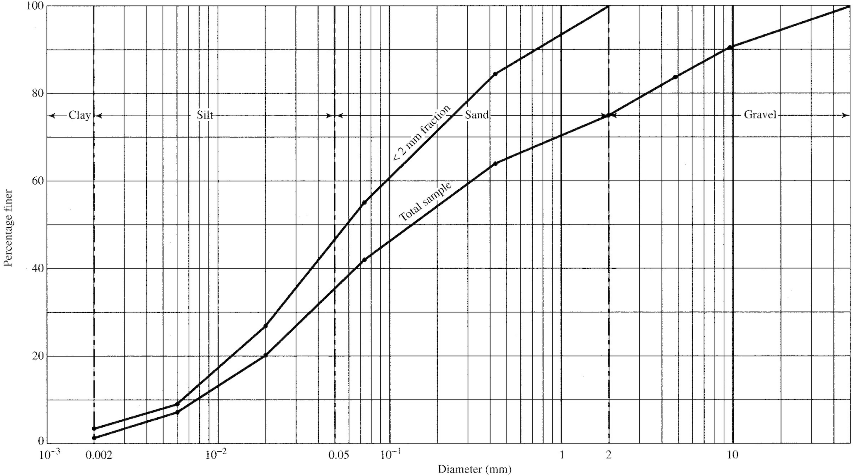


Grain Sizes

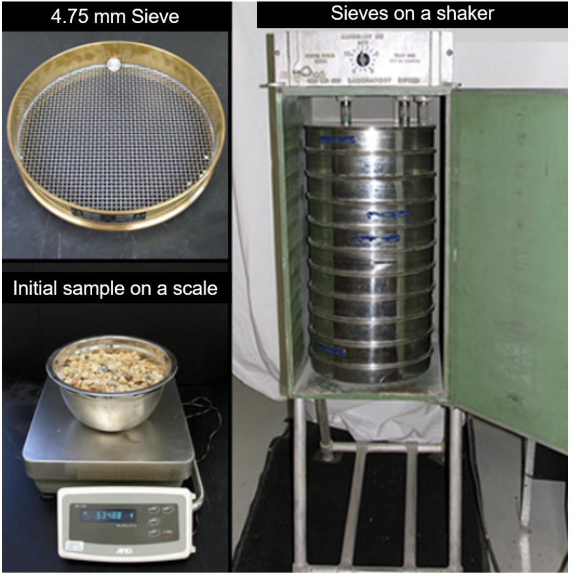


Grain size distribution

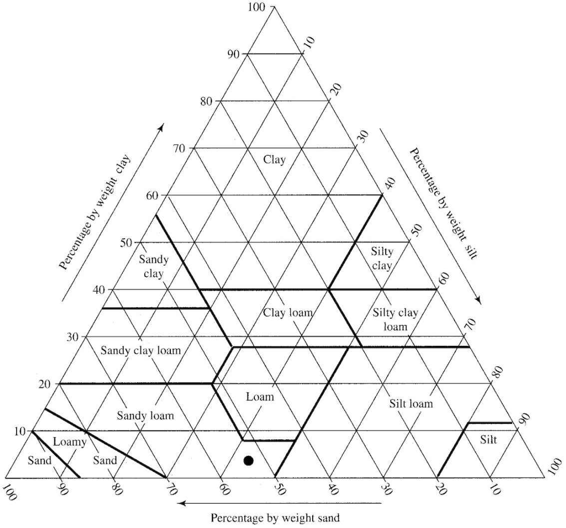
Cumulative-frequency plot of log-scale grain diameter. Steeper slope means more uniform.



Grain size distribution is determined by sieve analysis. Sample is dried, weighed, and then placed at the top of a stack of sieves (increasingly smaller mesh), shaken, and weighed.



Soil Texture



Weight proportions of clay, silt, and sand after grains $>$ 2 mm (e.g., gravel) have been removed.

Particle density

Weighted average of the density of particles making up the soil.

$$\rho_m = \frac{M_m}{V_m}$$

where M_m and V_m is the mass and volume of mineral grains, respectively.

Measurement: we just assume quartz because silica makes up most sediment, 2.65 g m^{-3}

Bulk density

Dry density of the soil.

$$\rho_b = \frac{M_m}{V_s} = \frac{M_m}{V_a + V_w + V_m}$$

where the *Vw* and *Va* is the volume of water and air, respectively.

Measurement: weigh soil after it has been dried (105 C for 16 hrs) and divide by initial bulk volume.

Porosity

Proportion of pore space in a porous media volume.

$$\phi = \frac{V_a + V_w}{V_s}$$

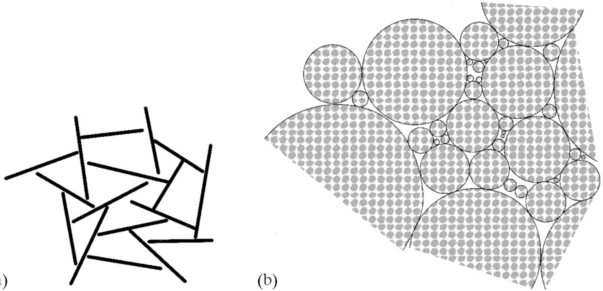
Measurement: it can be shown that:

$$\phi = 1 - \frac{\rho_b}{\rho_m}$$

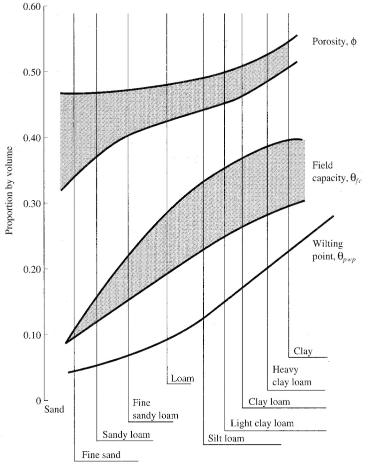
so ϕ can be estimated by measuring ρ_b and assuming a value for ρ_m

Porosity

Porosity is higher for finer-grained than coarser-grained porous media due to the "house of cards" arrangement of clay grains.



(a) (b)



Volumetric Water Content

aka soil moisture. Expressed as VWC or θ. Ratio of water volume to porous media volume. Can vary (in space and time) from 0 (dry) to ϕ (saturation)

$$\theta = \frac{V_w}{V_{pm}}$$

Lab measurement: weigh and measure wet sample, dry and reweigh:

$$\theta = \frac{M_{pm,wet} - M_{pm,dry}}{\rho_w \cdot V_{pm}}$$

A few more terms...

Degree of saturation (S) or wetness. Proportion of pores that contain water.

$$S = \frac{V_w}{V_a + V_w} = \frac{\theta}{\phi}$$

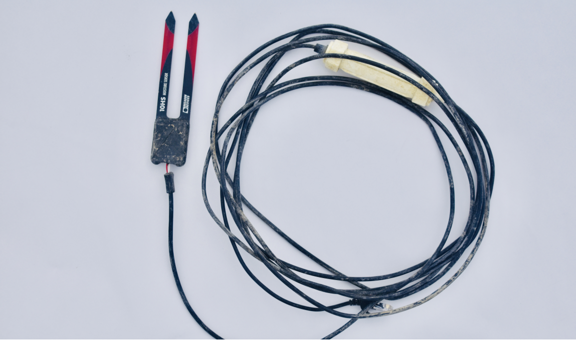
Total vadose storage, units of L, θ times thickness.

Measuring VWC in the Field

1. Capacitance
2. Time-domain reflectometry
3. Neutron probe
4. Electrical resistance
5. Remote sensing

Capacitance

Measures VWC indirectly via the dieletric constant (ε), which represents the ability of porous medium to store energy. Higher VWC = higher ε.



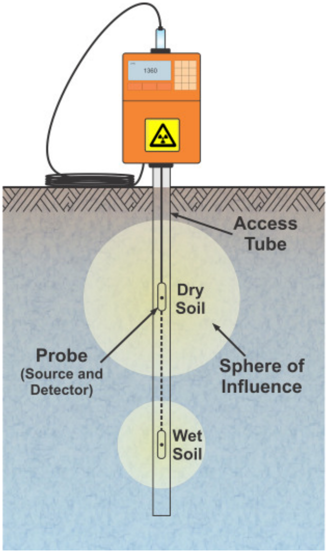
Most affordable in situ measurement, but requires site-specific calibration and is affected by soil conditions (salinity, clay content, temperature, bulk density).

Time-domain Reflectometry (TDR)

Measures VWC indirectly via the dieletric constant (ε), which represents the ability of porous medium to store energy. Higher VWC = higher ε.

More expensive than capacitance, very accurate (not impacted by soil conditions), very small area of measurement.

Neutron Probe



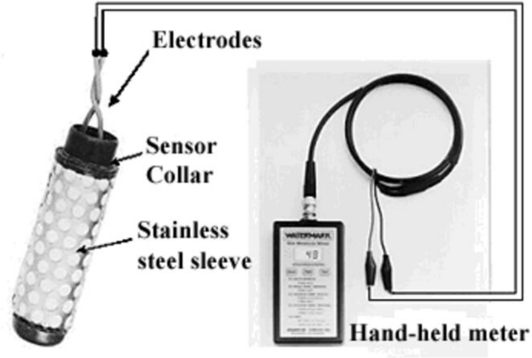
Sensor lowered down tube and emits high-energy neutrons (Americium 241/Beryllium) whose speed is attenuated when collided into hydrogen atoms.

Rate of attenuation $\propto$ VWC

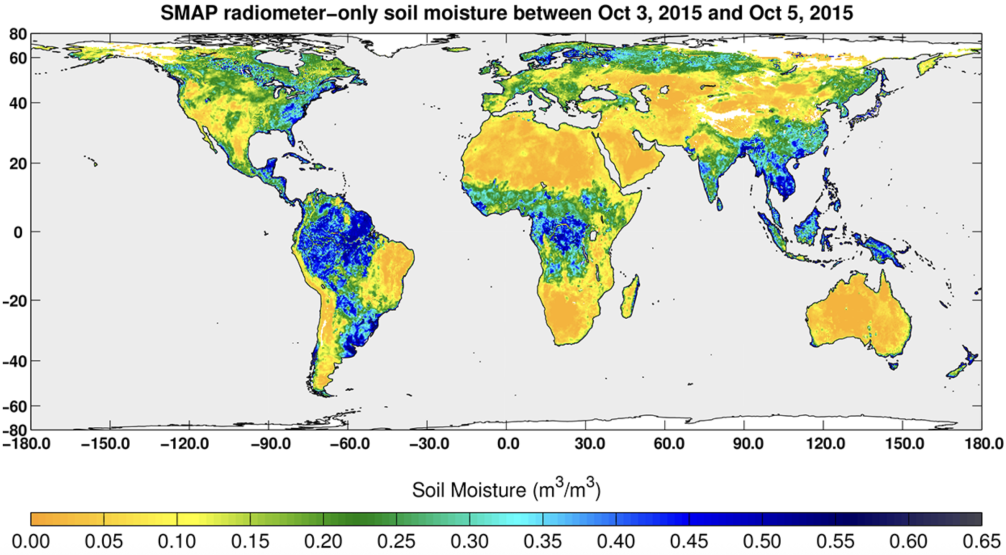
Very expensive, most accurate and works for a relatively large area. However, it contains radioactive material, requires special training/licensing, and is labor intensive.

Electrical Resistance Blocks

aka gypsum blocks. Use inverse relationship between water content and electrical resistance of porous medium (gypsum block) in equilibrium with the soil.



Remote Sensing



Vadose Zone Flow

Infiltration and redistribution in porous media can be described by Darcy's Law:

$$q_x = -K_h \cdot \frac{d(z + \frac{p}{\gamma_w})}{dx}$$

where q_x is the volumetric flow rate in the x-direction per unit cross-sectional area (LT^{-1}), K_h is hydraulic conductivity (LT^{-1}), z is the elevation above an arbitrary datum (sea level) (L), p is the water pressure ($ML^{-1}T^{-2}$), and γ_w is the weight density of water ($ML^{-2}T^{-2}$) calculated as:

$$\gamma_w = \rho_w \cdot g$$

Darcy's Law for 1-D, vertical flow

For unsaturated flow, p $\leq$ 0, so water flows from high (less negative) to low (more negative).

$$q_x = -K_h \cdot \frac{d(z + \frac{p}{\gamma_w})}{dx}$$

$$q_h = -K_h \cdot \left(\frac{dz}{dx} + \frac{d(p/\gamma_w)}{dx} \right)$$

$\cancel{dx}$ dz $\qquad$ dz

$$q_z = -K \cdot \left(1 + \frac{d(p/\gamma_w)}{dz} \right)$$

Darcy's Law for 1-D, vertical, unsaturated flow

$$q_z = -K_h \cdot [1 + \frac{d(p/\gamma_w)}{dz}]$$

weight density is effectively constant (barring major variations in salinity), so we can replace it with pressure head (ψ) with units of L:

$$\psi = \frac{p}{\gamma_w}$$

both hydraulic conductivity and ψ are dependent on water content (θ):

$$q_z = -K_h(\theta) \cdot \left[1 + \frac{d\psi(\theta)}{dz}\right]$$

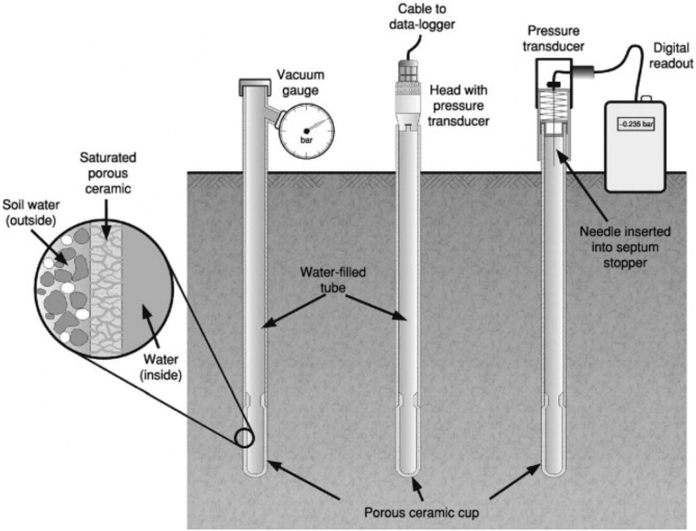
Notes on Pressure in the Vadose Zone

- Pressure, (F/A, with units of $ML^{-1}T^{-2}$ or FL^{-2}) is a scalar quantity acting in all directions of a fluid.
- p and $\psi > 0$ in saturated flows and < 0 in unsaturated flows.
- p = 0 at the water table.
- negative p is called tension/suction
- ψ (pressure head) is called tension head, matric potential, or matric suction when p < 0
- tension increases (p gets more negative) when water content decreases

Measuring Tension in the Field

Tension can be directly measured using **tensiometers.**

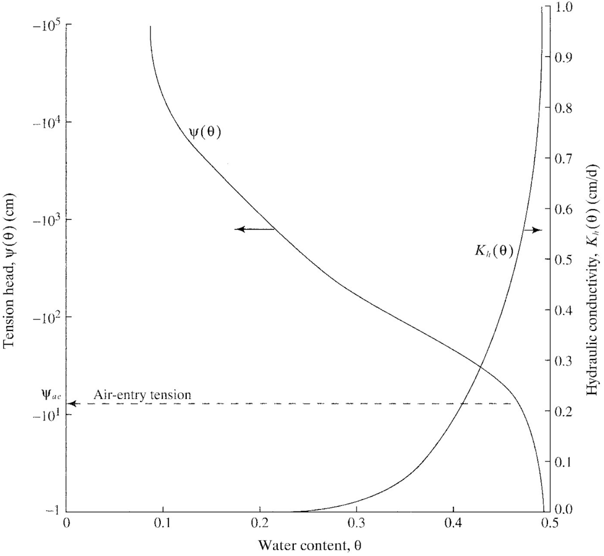
$$|\psi| = |\psi_{gage}| - z_{ten}$$



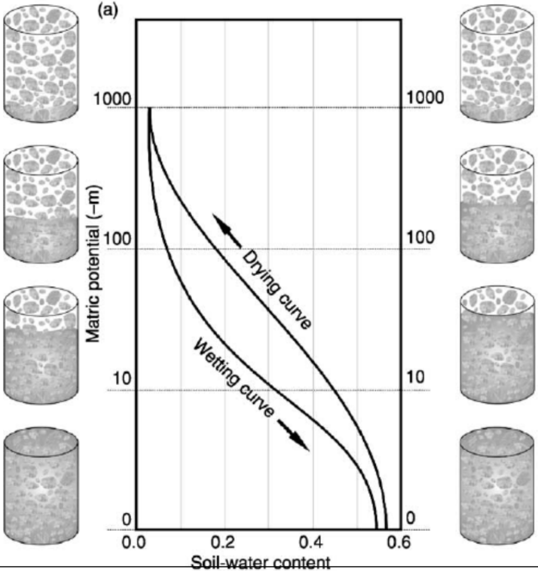
Pressure-Water Content Relations

"Moisture characteristic curve": nonlinear relationship between ψ and θ.

Minimal change in θ until an inflection point is reached, where significant amounts of air appear in soil pores. This is the **air entry tension** (ψ_{ae}).



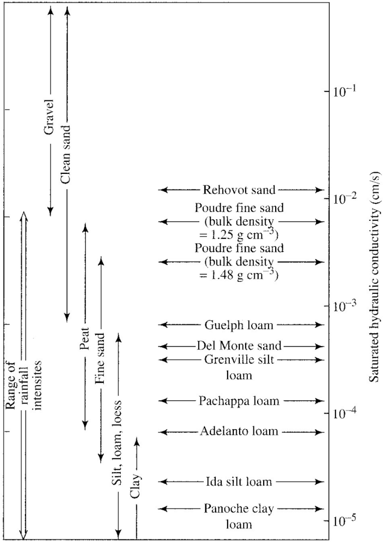
Hysteresis

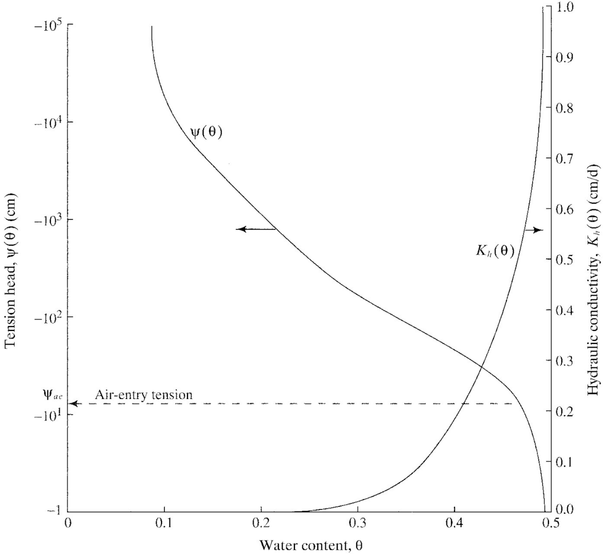


The moisture characteristic curve is **hysteretic**; it differs for wetting versus draining, where the porous media will have higher water content at the same matric potential when draining.

Reasons include: entrapped air and the "ink bottle" effect

Hydraulic Conductivity-Water Content Relations





Field Capacity

The water content below which further decrease in further drainage occurs at a negligible rate. It reflects the water content that can be held against gravity but still accessible to plants.

$$\theta_{fc} = \phi \cdot (\frac{|\psi_{ae}|}{340})^{1/b}$$

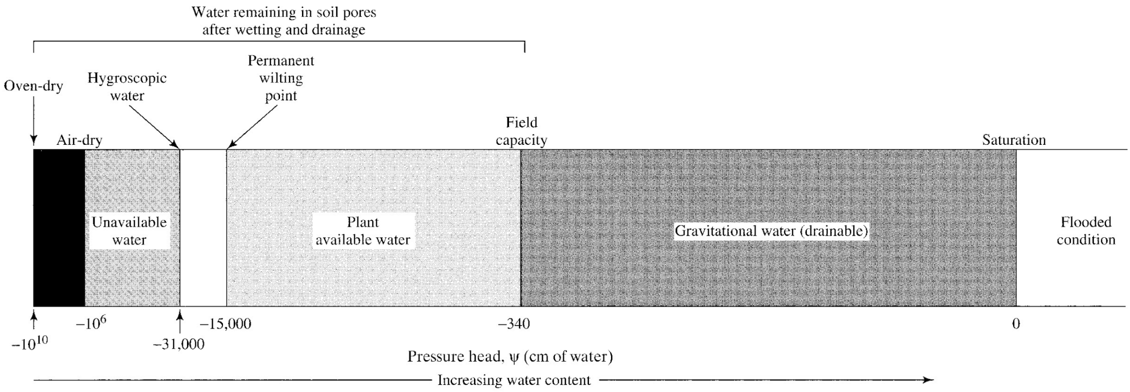
where θ_{fc} is the field capacity, ϕ is porosity, ψ_{ae} is the air entry tension (cm), and b is the exponent describing the moisture characteristic curve (from a table).

Permanent Wilting Point

Plants cannot exert tension greater than about -15,000 cm (-1470 kPa), so when θ falls low enough that ψ is below that, transpiration ceases and the plant wilts.

$$\theta_{pwp} = \phi \cdot (\frac{|\psi_{ae}|}{15,000})^{1/b}$$

The difference between θ_{fc} and θ_{pwp} is the **available water content.**


Water remaining in soil pores
after wetting and drainage
Oven-dry
Hygroscopic
water
Permanent
wilting
point
Field
capacity
Saturation
Air-dry
Unavailable
water
Plant
available water
Gravitational water (drainable)
Flooded
condition
-10^{10}
-10^6
−31,000
−15,000
−340
0
Pressure head, ψ (cm of water)
Increasing water content